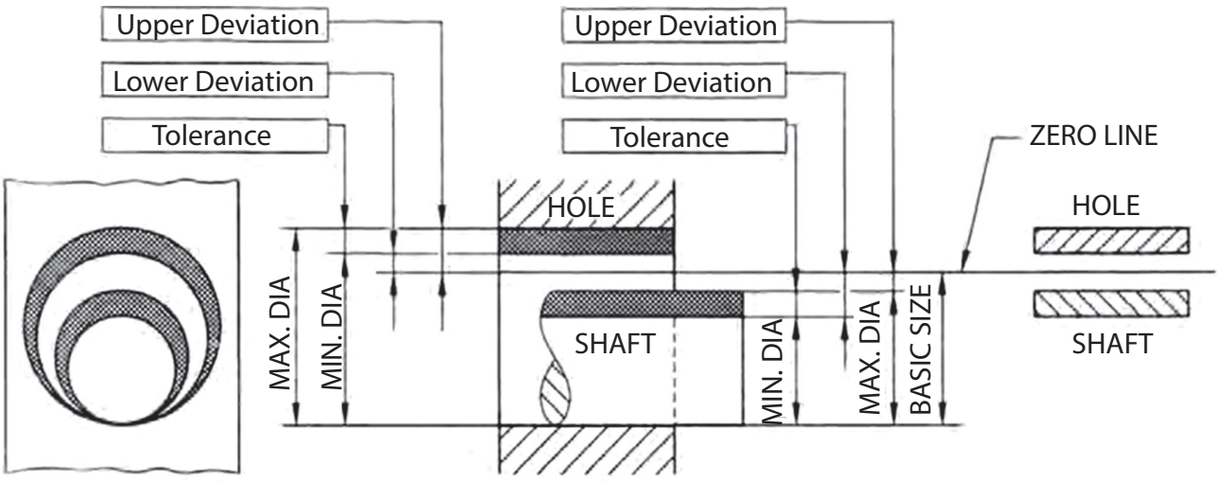


Basic Terminology In Interchangeble System

represents any external dimension of a component.

Hole: The hole indicates the inner diameter of a cylinderical hole but also represent any internal dimensions of a component.

Basic Size: Basic size of a dimension is the size in relation to which all limits of variations are determined. This is fixed up by designer considering its functional aspects without indicating any tolerance.

Actual Size: It is defined as the size of actually obtained by machining. It is found by measurement using measuring instruments.

Limits of Size: Limits are the two extreme permissible sizes of any dimension, the actual size should lies between these two limits of sizes.

Deviation: The algebraic difference between the actual size and its corresponding basic size is called deviation.

Upper Deviation: It is defined as the algebraic difference between the maximum limit of size and the corresponding basic size.

Lower Deviation: It is defined as the algebraic difference between the minimum limit of size and the corresponding basic size.

Zero Line: The deviations are always measured from basic size. Therefore, to represent limits and fits graphically, a straight line is drawn for basic size. This line is called zeroline because the deviation at the basic size is zero.

When zero line is drawn horizontally deviations above this line will be positive and below it will be negative. The sign + is added with positive and the sign – is added with negative deviations.

TOLERANCE

Tolerance is the difference between the maximum limit of size and minimum limit of size.

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